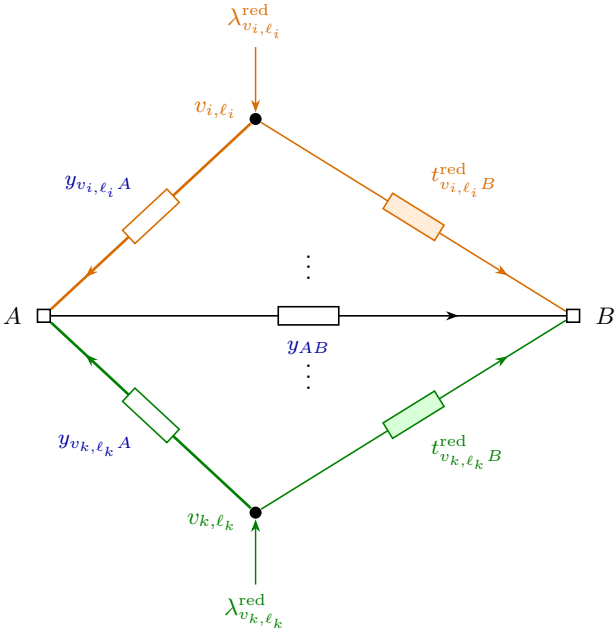


$y_{v_{i,\ell_i}A} = r_{i,2\ell_i-1}\lambda^{\mathrm{red}}_{v_{i,\ell_i}} - r_{i,2\ell_i}\, y^{\mathrm{red}}_{AB}$

$y^{\mathrm{red}}_{AB} = \lambda_A + \sum_k r_{k,2\ell_k-1}\lambda^{\mathrm{red}}_{v_{k,\ell_k}}$

the arms couple only through y^{red}_{AB}

rows of $y_{v_{i,\ell_i}A}$:

column $r_{k,2\ell_k-1}$: $(\delta_{ik} - r_{i,2\ell_i})\,\lambda^{\mathrm{red}}_{v_{k,\ell_k}}$

column $r_{i,2\ell_i}$: $-\,y^{\mathrm{red}}_{AB}$